

Satheesh R

SUMMARY

Full Stack Engineer with strong expertise in the **MERN** stack, capable of building scalable, user-focused applications across the entire development lifecycle. Experienced in crafting responsive UIs, secure APIs, and cloud-deployed solutions. Known for delivering maintainable and impactful products across varied domains.

SKILLS

Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js, Angular, Vue.js, HTML5, CSS3, Tailwind CSS, Framer Motion

Backend: Node.js (Express), Django, RESTful APIs

Databases: MongoDB, MySQL, Redis, PostgreSQL

DevOps & Deployment: Vercel, Netlify, GitHub, AWS Services, Bitbucket, Azure

CMS & Platforms: WordPress, Webflow, Webiny CMS

Tools & Project Management: Git, Jira

Other: Basic Networking (TCP/IP, Routing), JWT Auth, CSRF, MetaMask

PROFESSIONAL EXPERIENCE

Securenext Softwares

Aug' 25 – present

Software Developer

Rag Bot(AI - Content-Retrieval Chatbot)

- Built a PDF-driven **RAG system** using **React**, **TypeScript**, **Python**, and **MongoDB** to let users upload documents; this allowed the model to search an average of 150+ pages per file, giving users direct answers pulled from the source with over **90% retrieval accuracy** based on the custom scoring metric implemented..
- Set up the local **Ollama Qwen 3.4** model and a custom retrieval prompt in Python to reduce irrelevant outputs by about **40%** while keeping all document data fully on-device for privacy.
- Used tensor operations to turn document chunks into embeddings for similarity search, reducing lookup time from **~2 seconds** to **~300 ms** and making question-answer retrieval noticeably faster.
- Added a voice system using **Tortoise-TTS** and **Hugging Face** models to **clone a user's voice** and read out generated answers; this let users consume responses **30–40%** quicker than reading long text blocks.
- Implemented text and **audio streaming** from the backend so responses begin appearing instantly, which lowered perceived wait time by over 50% and made the interaction feel more real-time for users.

AI Health tech:

- Built a workflow using **React**, **Python**, **PostgreSQL**, and **Azure** to generate **AI-driven content**, processed the ZIP server-side to pull out only the needed file, and delivered that file to the **Slate editor** so users could work with clean, focused data.
- Set up a flow where curated **AI output** is loaded into the Slate editor to let users revise the content, and then saved the updated file to cloud storage to keep a clear record of each version.
- Designed a backend process that pulls the previously edited file, filters it to show only the sections needing attention, and lets users make further changes without overwriting their earlier edits.

Clar Technologies

Apr' 25 – July' 25

Doctor's CRM(AI - CRM for Doctors)

- Carefully implemented an **AI-powered diagnosis** system using a fine-tuned **CheXNet model** in Django to accurately detect pneumonia from X-ray images uploaded through local storage, enabling doctors to receive diagnostic results promptly within the CRM.
- Systematically developed a full-stack application using **Django**, **HTML**, and **CSS**, providing hospitals with a clear and organized admin panel to manage doctor schedules, review diagnostic reports, and maintain patient medical history.
- Securely integrated **CSRF protection** and **JWT-based authentication** to control user access and safeguard sensitive patient data, following strict medical data privacy practices. Blockchain Betting Game
- Engineered a decentralized betting game using **React** and smart contract interactions on **ApeChain** and **Abstract Blockchain**, allowing users to participate in **NFT-based** wagers transparently.
- Integrated **MetaMask** for wallet connection, enabling secure login and **blockchain-based transaction** execution; users can place bets and receive outcomes directly on-chain.
- Fetched and validated real-time **NFT metadata** from blockchain to authorize gameplay, while updating the UI dynamically to reflect transaction status and enhance gameplay experience.

Edufied Pte Ltd	Jan '24 – Jul '24
Front End Developer Trainee, Singapore	
• Crafted customizable certificate templates and spearheaded frontend using Angular, transforming customization and improving user experience by 35% • Resolved 10+ critical frontend issues in a 50,000+ line Angular/TypeScript project, strengthening stability and accessibility • Revamped UI performance and responsiveness, boosting user engagement by 25% while managing code with Bitbucket and tracking 30+ tasks via Jira • Constructed a no-code website builder using Webiny CMS, React, and TypeScript, empowering custom website creation with AWS deployment for 3x scalability	
Lumos Ind	Aug '20 – Jan '21
Research Associate, Coimbatore, Ind	
• Researched AR/MR/VR technologies (HoloLens, Meta Quest) and authored a 30-page report with 12+ case studies. Identified 8+ COVIDera use cases in healthcare, education, and emergency response.	

PROJECTS

Website Builder – Drag & Drop Page Creator	Mar '25
• Built a no code website builder using Next.js, React, TypeScript, and Tailwind CSS, enabling users to design responsive sites visually without writing code • Implemented drag-and-drop functionality with @hello-pangea/dnd and a property editor for customizing element styles, layout, and structure • Integrated HTML/CSS export, responsive preview modes, and local template saving to enhance usability and deployment flexibility • Designed a modular architecture with reusable components and modern UI via shadcn/ui, improving code maintainability and scalability	
Genius (AI-SaaS)	Apr '24
• Architected an AI-powered SaaS platform for 5 content types(text, code, image, music, video)using Next.js, React & Tailwind CSS • Designed a fully responsive landing page, streamlining user onboarding by 40% by showcasing platform features and eliminating access barriers • Integrated secure Google authentication using Clerk, enabling seamless account management and 2-step sign-in processes • Launched an interactive user dashboard incorporating Riffusion for music generation, Zeroscope for video creation, and OpenAI models for code generation • Established a real-time chat system using Crisp, facilitating instant developer communication and reducing issue resolution time by 65%	
Archadius (Portfolio)	Oct '24
• Programmed a fully responsive and interactive web application using React, securing cross-browser compatibility and optimal performance across 7+ devices • Executed advanced code-splitting techniques, decreasing load times by 50% and minimizing resource consumption	

EDUCATION CREDENTIAL

Amrita Vishwa Vidyapeetham , Coimbatore, Ind	2024
Masters in Computer Application	
Dr G R Damodaran College of Science , Coimbatore, Ind	2021
Bachelors of Information Technology	